

EXPERIMENT - 3

Design a Virtual LAN for LAB 1.a AND LAB 2.b

AIM :

To design and configure a Virtual Local Area Network (VLAN) for AIML and CSD departments located in Block-A and Block-B using switches, such that faculty members of the same department are logically grouped in the same LAN while maintaining separate broadcast domains, and to verify connectivity using PDU packets in Cisco Packet Tracer.

PROCEDURE :

1. Network Design

1. Open Cisco Packet Tracer.
 2. Place two switches representing Block-A and Block-B.
 3. Add PCs for faculty members:
 - AIML faculty PCs in both blocks
 - CSD faculty PCs in both blocks
 4. Connect PCs to their respective switches using Copper Straight-Through cables.
 5. Connect Switch-A and Switch-B using a Copper Cross-Over / Straight-Through cable (auto-MDIX supported).
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2. VLAN Creation

1. Click on Switch-A → CLI.
2. Repeat the same VLAN creation steps on Switch-B.

3. Assigning VLANs to Ports

Assign AIML faculty ports to VLAN 10 and CSD faculty ports to VLAN 20.

Repeat the same steps for the corresponding ports on Switch-B.

4. Trunk Configuration Between Switches

1. Configure the inter-switch link as a trunk on both switches.

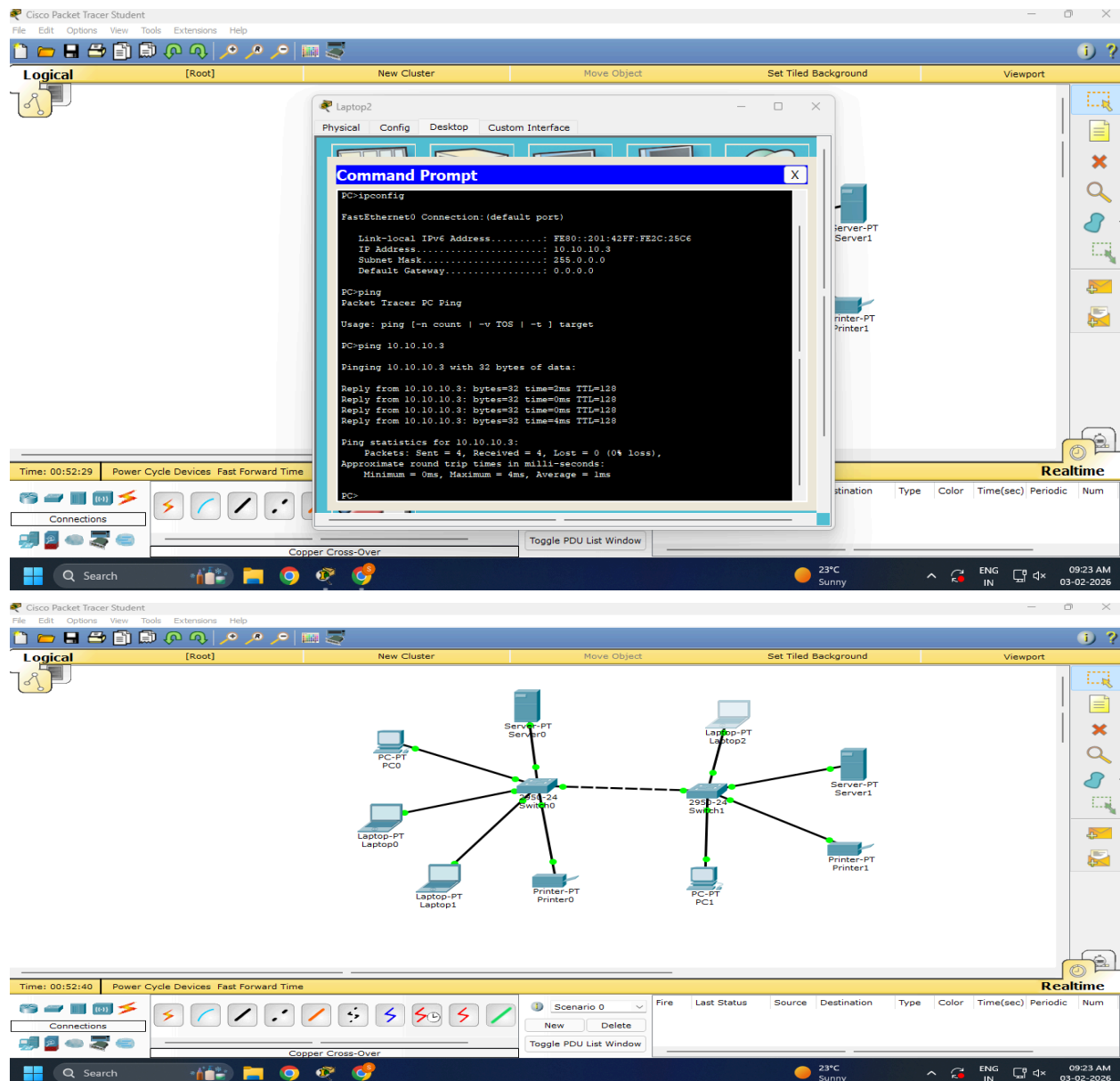
5. IP Address Configuration

Assign IP addresses from the same network for each VLAN.

6. Connectivity Testing

1. Switch to Simulation Mode.
2. Use Add Simple PDU.
3. Send PDU:
 - Between AIML PCs in Block-A and Block-B → Successful
 - Between CSD PCs in Block-A and Block-B → Successful
 - Between AIML and CSD PCs → Failed (different broadcast domains)

OUTPUT :



RESULT :

A LAN was successfully created using a Switch-PT and five PCs. All systems were assigned Class A IP addresses with the subnet mask 255.0.0.0, placing them in the same subnet. Communication between all PCs was verified successfully using the ping command.

